0.1 Theoretical Model

Consider an environment with a two-tiered production network. Downstream firms, indexed by j, sell differentiated varieties to consumers. Upstream firms, indexed by f, sell inputs used in production by the downstream firms. Firms, both upstream and downstream, will be able to alter their productivity through their choice of R&D, represented by $\boldsymbol{I}$, or production structure, represented by $\boldsymbol{\omega}$. Aggregate R&D, $\boldsymbol{\mathcal{I}}$, and aggregate network structure, $\boldsymbol{\Omega}$, will depend on firm-level choices and additionally influence production technology. Factor prices are determined by a freely-traded outside good.[1]

Technology. I specify technology in terms of a upstream and downstream firm's cost of production:

$$c_{f,j}\Big(\boldsymbol{I}_f,\boldsymbol{\omega}_f,\boldsymbol{\mathcal{I}},\boldsymbol{\Omega}\Big) \text{ and } c_j\Big(\bar{\boldsymbol{p}}_{j,I},\boldsymbol{I}_j,\boldsymbol{\omega}_j,\boldsymbol{\mathcal{I}},\boldsymbol{\Omega}\Big).$$

Each firm can affect productivity through R&D expenditures $\{\boldsymbol{I}_j,\boldsymbol{I}_f\}$ - potentially along a number of input-specific, buyer-specific, or country-specific dimensions. Each firm chooses a production structure that alters costs, $\{\boldsymbol{\omega}_j,\boldsymbol{\omega}_f\}$. As an example, this may represent locations decisions - both the choice of where to produce and where to do R&D - or firm-to-firm linkages made between suppliers and buyers. Downstream costs additionally depend on a vector of input prices, $\bar{\boldsymbol{p}}_{j,I}=\{\bar{p}_{f,j}\}_f$, provided by upstream firms.[2]

The aggregates state - which includes moments of R&D expenditure, $\boldsymbol{\mathcal{I}}$, and industry structure, $\boldsymbol{\Omega}$ - additionally affect costs. The state variable $\boldsymbol{\mathcal{I}}$ represents a standard form of Marshallian externalities from collective knowledge production, while aggregate industry $\boldsymbol{\Omega}$ structure capture other forms of spillovers like geographic agglomeration forces. These aggregates can be multi-dimensional and depend differently on the components of R&D and production structure.[3]

Policy. A planner will have the opportunity to affect the prices and costs enter the firms' problems. In particular, I write final prices $\{\bar{p}_j\}$, input prices $\{\bar{p}_{f,j}\}$, and R&D costs $\{\bar{r}_{j,k},\bar{r}_{f,l}\}$ as a function of the underlying variables and policy wedges:

$$\begin{aligned}
\bar{p}_j &= (1-t_j)\,p_j,\\
\bar{p}_{f,j} &= (1-t_f)\,(1-t_{f,j})\,p_{f,j},\\
\bar{r}_{j,k} &= (1-s_{j,k})\,r,\\
\bar{r}_{f,l} &= (1-s_{f,l})\,r.
\end{aligned}$$

The indices k and ℓ are for each separate dimension of $\boldsymbol{I}_j$ and $\boldsymbol{I}_f$.[4] In addition, I assume that the cost of a particular production structure takes the following form

$$\bar{f}\,(\omega_f)=f\,(\omega_f)+\phi\,(\omega_f) \text{ and } \bar{f}\,(\omega_j)=f\,(\omega_j)+\phi\,(\omega_j)$$

[1] I interpret the model results as characterizing optimal policy for one sector, sufficiently small that outcome on this sector do not affect economy-wide aggregates like factor prices or preferences.

[2] The dependence of costs on factor prices is suppressed into the choice of production structure ω_j,ω_f. For instance, ω may denote the location of production. If locations vary only in the cost of labor, differences in cost under various values of ω simply reflects difference in wages.

[3] For example, there may be country-specific aggregate R&D levels that depend only on R&D spending done in those countries.

[4] For example, $\boldsymbol{I}_j$ and $\boldsymbol{I}_f$ may consist of location-specific R&D components that differentially affect cost according to xxx.

where $\{\phi^U(\omega_j), \phi^D(\omega_j)\}$is a tax/subsidy associated with production structure ω_j.

Profit Maximization. Downstream firms determine final prices, R&D expenditures, and firm structure to maximize profits:

$$\pi_j = \max_{\bar{p}_j, \boldsymbol{I}_j, \omega_j} q_j\left(p_j\right)\left(\bar{p}_j - c_j\left(\bar{\boldsymbol{p}}_{j,I}, \boldsymbol{I_j}, \omega_j, \boldsymbol{\mathcal{I}}, \boldsymbol{\Omega}\right)\right) - \bar{\boldsymbol{r}}_j \boldsymbol{I}_j - \bar{f}\left(\omega_j\right). \tag{1}$$

Downstream firms each face a residual demand curve and wield some monopoly power. I assume that R&D resources are in perfectly elastic supply at a cost $\bar{\boldsymbol{r}}_j$ to the firm. Finally, each production structure is associated with fixed cost payment.

Likewise, upstream firms face a similar profit maximization problem:

$$\pi_f = \max_{\left\{\bar{p}_{f,j}\right\}_j, \boldsymbol{I}_f, \omega_f} \sum_j q_{f,j}\left(p_{f,j}\right)\left[\bar{p}_{f,j} - c_{f,j}\Big(\boldsymbol{I}_f, \omega_f, \boldsymbol{\mathcal{I}}, \boldsymbol{\Omega}\Big)\right] - \bar{\boldsymbol{r}}_f \boldsymbol{I}_f - \bar{f}\left(\omega_f\right) \tag{2}$$

where the substantive difference of note is that profits come from the sum of sales across all buyers.

Aggregate State. The firm-level choices of R&D expenditure $\{\boldsymbol{I}_f, \boldsymbol{I}_j\}$ and structure $\{\omega_f, \omega_j\}$ determine the aggregate R&D levels and industry structure. And these aggregate can be multi-dimensional functions of the firms' choices:

$$\boldsymbol{\mathcal{I}} = \boldsymbol{\mathcal{I}}\left(\{\boldsymbol{I}_f, \boldsymbol{I}_j\}\right) \text{ and } \boldsymbol{\Omega} = \boldsymbol{\Omega}\left(\{\omega_f, \omega_j\}\right). \tag{3}$$

[X] What if $\boldsymbol{\mathcal{I}} = \boldsymbol{\mathcal{I}}\left(\{\boldsymbol{I}_f, \boldsymbol{I}_j\}, \{\omega_f, \omega_j\}\right)$? So the local R&D spillovers are directly effects by production structure?

Utility Maximization. A representative agent chooses an optimal consumption profile over the varieties produces by downstream firms and an outside good C

$$U = \max_{C, \{c_j\}_j} u\left(C, \{c_j\}_j\right) \tag{4}$$

subject to $C + \sum p_j c_j \leq \boldsymbol{wL'} + \sum_f \pi_f + \sum_j \pi_j + T$. I set as numeraire the price of the outside good C, use $\boldsymbol{wL'}$ to denote the total payments received by factors (including payments for R&D and entry), and T to denote the net government revenue from taxes and subsidies.

Equilibrium. An equilibrium is defined by a set of prices, R&D investments, and production structures that solve 1, 2, and 4 - subject to 3 as well as standard market clearing and budget constraints. Note, this model may admits the possibility of multiple equilibria depending on the extent of complementarities in R&D and network spillovers. Policy has the opportunity to improve outcomes either by fixing allocations within a given equilibrium, or by directing firms towards selecting an equilibrium with higher aggregate welfare.

0.1.1 Optimal Policy

A planner's objective is to choose a set of policy instruments, $\{t_j, t_f, t_{f,j}, s_{j,k}, s_{f,l}, \phi(\omega_f), \phi(\omega_j)\}$, that maximize welfare according to 4. I characterize some features of the optimal industry policy

Proposition 1. *Assume the choice of production structure is fixed. A planner chooses a standard set of taxes on final production and supplier production that corrects the monopoly distortion. They choose a set of taxes on the purchase of inputs that satisfies*

$$1 - t_{f,j} = \frac{q_j}{q_{f,j}} \frac{\partial c_j}{\partial p_{f,j}}$$

and they choose an R&D subsidy that satisfies

$$s_f = \sum_j \frac{\partial c_j}{\partial \boldsymbol{\mathcal{I}}} \frac{\partial \boldsymbol{\mathcal{I}}}{\partial \boldsymbol{I}_f} q_j + \sum_{f,j} \frac{\partial c_{f,j}}{\partial \boldsymbol{\mathcal{I}}} \frac{\partial \boldsymbol{\mathcal{I}}}{\partial \boldsymbol{I}_f} q_{f,j}.$$

The first tax corrects the extent to which upstream firms do not fully internalizes the benefits of price-reductions to downstream firms. If Shephard's lemma held for the cost function implied by 1, then the first tax would not be necessary.[5] However, in the presence of an endogenous R&D allocation and thus potential non-constant returns to scale, a reduction in the price of inputs can lead downstream firms to change their R&D expenditure and affect their unit cost - an effect no internalized by suppliers. The second tax corrects for the extent to which all firms do not fully internalize how R&D investment spills over into the aggregate R&D state $\boldsymbol{\mathcal{I}}$.

The next set of policies regards correcting the choice of ω_j. I can not characterize a fully generic policy here, so I specialize the model to a simplified case with only two types of upstream and downstream firms. As an intuitive example, one can think of these choices are representing the choice of suppliers or downstream firms to offshore production.

To make sense of what policy may look like for production structures, consider the following case. There is a unit mass of homogenous upstream and downstream firms. Firms can choose one of two structures $\omega_f \in \{0, 1\}$ and $\omega_j \in \{0, 1\}$ such that the mass of type 1 upstream and downstream firms can be represented by M and N. Suppose that the cost function is monotonic in all arguments. One can heuristically think about these as extensive margin choices to offshore production.

Proposition 2. *Consider the case described in the previous paragraph. Suppose an equilibrium allocation is interior, so that $M, N \in (0, 1)$, and marginal changes in M and N are well defined. Then, the optimal tax on type 1 firms is*

$$\left(\sum_j \frac{\partial c_j}{\partial \boldsymbol{\Omega}} q_j + \sum_{f,j} \frac{\partial c_{f,j}}{\partial \boldsymbol{\Omega}} q_{f,j} \right) \times \frac{d\Omega}{dM} + \left(\sum_j \frac{\partial c_j}{\partial \boldsymbol{\mathcal{I}}} q_j + \sum_{f,j} \frac{\partial c_{f,j}}{\partial \boldsymbol{\mathcal{I}}} q_{f,j} \right) \times \left(\boldsymbol{I}_{f|\omega_f=1} - \boldsymbol{I}_{f|\omega_f=0} \right) = \phi_f.$$

A tax is required to correct two externalities through the aggregate states. First, firms do not internalize how their choice to offshore affects the aggregate industry structure which alters the unit cost of other firms.

[5] Solving 1 for the optimal allocation of R&D implies a cost-function that can be written as a function of factor prices and quantities. See Appendix Section x for an example. If Shephard's lemma held, then $q_j \frac{\partial c_j}{\partial p_{f,j}} = q_{f,j}$ and the implied tax is 0.

For instance, if there are geographic agglomeration affects from the proximity of manufacturing. Second, firms do no internalize that the different R&D levels that come with offshoring will affect other firms through the Marshallian externalities. For example, if offshoring is a substitute for R&D or causes firms to relocate R&D abroad.[6]

[X] Second-Best Policy Instruments, [X] Clean Proofs, [X] Strategic Considerations: prop4_offshoring_pd.lyx [X] Strategic Policy Considerations

[6] I model R&D as a productivity-improving endeavor. As a result, the marginal returns to R&D are high when costs are high. However, high costs also tend to lower quantities sold, and the marginal returns to R&D rise with quantities. In order for offshoring to be a substitute for R&D, the cost reduction has to offset the quantity gains.

0.2 Parametric Model

In order to take the model to the data, formalize additional theoretical results, and make precise quantitative statements, I need to parametrize the model. This involves taking stances on the functional forms for costs and preferences. This section implements a simplified model for applied work, but in the later sections I enrich the model further for an empirical implementation.

Technology. I start by specifying the cost of production for downstream firms:

$$c_j = \frac{\bar{c}_{j,A}}{I_{j,A}^{\eta}} + \frac{p_{j,I}}{I_{j,I}^{\eta}} \tag{5}$$

where $\bar{c}_{j,A}$ is the cost of assembly, $p_{j,I}$ is the price of inputs, and $I_{j,A}, I_{j,I}$ represent R&D on the components of production.[7] Total R&D is a aggregation of country-level R&D expenditures, where countries differ in their R&D productivity either due to the stock of local knowledge $\mathcal{I}_c$ or manufacturing proximity effects:

$$I_{j,A} = \left(\sum_{c \in C_j} \left[I_{j,A,c} \zeta_{f,A,c} \mathcal{I}_c^{\omega} d_{j,A,c}^{\tau_j} \right]^{\varsigma} \right)^{1/\varsigma} \text{ and } I_{j,I} = \left(\sum_{c \in C_j} \left[I_{j,I,c} \zeta_{f,I,c} \mathcal{I}_c^{\omega} d_{j,I,c}^{\tau_{jf}} \right]^{\zeta} \right)^{1/\varsigma} \tag{6}$$

where $d_{j,A,c}$ is the proximity of firm j to its own assembly plants while $d_{j,I,c}$ is the proximity to its suppliers manufacturing plants. The summand is over the set of countries C_j that firm j operates R&D facilities in. Finally, the price of inputs is given by

$$p_{j,I}^{1-\theta} = \sum_{f \in M_j} (p_{f,j})^{1-\theta}$$

where M_j is the set of supplier faced by j.[8]

Next, for upstream firms, the cost of production is given by

$$c_{f,j} = \frac{\bar{c}_{f,j}}{I_f^{\rho} I_{f,j}^{\rho}}$$

where $\bar{c}_{f,j}$ is the baseline cost of production by supplier f for buyer j and $I_f, I_{f,j}$ represent a generic and buyer-specific (specialized) component of R&D. As before, R&D is a aggregation of country-level R&D expenditures where countries differ in their R&D productivity either due to the stock of local knowledge $\mathcal{I}_c$ or manufacturing proximity effects :

$$I_f = \left(\sum_{c \in C_f} \left[I_{f,c} \zeta_{f,c} \mathcal{I}_c^{\omega} d_{f,c}^{\tau_f} \right]^{\varsigma} \right)^{1/\varsigma} \text{ and } I_{j,I} = \left(\sum_{c \in C_f} \left[I_{f,j,c} \zeta_{f,j,c} \mathcal{I}_c^{\omega} d_{f,j,c}^{\tau_{fj}} \right]^{\zeta} \right)^{1/\varsigma} \tag{7}$$

where $d_{f,c}$ is the proximity of firm f to its own manufacturing plants while $d_{f,j,c}$ is the proximity to its

[7] An additive cost function implies that assembly and inputs are perfect complements in the downstream production function. In the empirical implementation I allow for imperfect substitution in the downstream production function, and estimation suggests inputs are complements. Derivations for a CES downstream production function between assembly and inputs are available in the Appendix.

[8] There are several ways to micro-found input demand of this firm. One interpretation is OEMs have a CES aggregation of supplier-specific varieties. The demand system for inputs specified in 10 follows from such a setup.

buyers manufacturing plants. The summand is over the set of countries C_f that firm f operates R&D facilities in.

In the notation of the theoretical model, the firm's structure can be summarized by the tuple $\boldsymbol{\omega}_j \equiv \{\bar{c}_{j,A}, \{d_{j,A,c}\}_c, C_j, M_j\}$ and for suppliers $\boldsymbol{\omega}_f \equiv \{\bar{c}_{f,j}, \{d_{j,f}\}_c, C_f\}$. A firms production structure determines their baseline production costs, their proximity to their own plants, their set of R&D facilities, and their set of suppliers. The R&D vectors are summarized by $\boldsymbol{I}_j \equiv \{I_{j,A,c}, I_{j,I,c}\}_c$ and $\boldsymbol{I}_f \equiv \{I_{f,c}, \{I_{f,j,c}\}_f\}_c$ - the country-level expenditures on assembly and input R&D as well as generic and specialized R&D. And finally, the aggregate states includes local R&D expenditure $\boldsymbol{\mathcal{I}} \equiv \{\mathcal{I}_c^{\omega}\}_c$ and the proximity to partner firms $\boldsymbol{\Omega} \equiv \{d_{j,I,c}, d_{f,j,c}\}$. XXX [partner plant locations determine part of $\bar{c}_{f,j}$]

Utility Maximization. In the baseline model, I consider CES demand for the differentiated varieties produced by downstream firms, subject to a fixed budget constraint:

$$\max_{x_j} \left(\sum_{j \in N} x_j^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \text{ subject to } \sum_{j \in N} x_j p_j \leq \bar{D}.$$

But in the Appendix Section X, I consider an extension to quasi-linear preferences where the aggregate industry price index determines the total level of demand.

Profit Maximization. Firm optimization unfolds over two stages. First, firms determine their production structure ω_j, ω_f. Second, R&D is allocated, input and final prices are determined, and goods are produced and consumed. I start by characterizing the second stage. In particular, the downstream firms choose final prices and R&D:

$$\pi_j(\boldsymbol{\omega}_j, \boldsymbol{\Omega}) = \max_{p_j, \{I_{j,A,c}, I_{j,I,c}\}_c} D(p_j)^{-\sigma} \left(p_j - c_j\left(\{p_{f,j}\}_f, \{I_{j,A,c}, I_{j,I,c}\}, \boldsymbol{\omega}_j, \boldsymbol{\mathcal{I}}, \boldsymbol{\Omega}\right)\right) - \sum_{c \in C_j} r_c\Big(I_{j,A,c} + I_{j,I,c}\Big) \quad (8)$$

where $D = \bar{D}P^{\sigma-1}$ and $P^{1-\sigma} = \sum_j p_j^{1-\sigma}$. Then, upstream firms determine R&D.

$$\pi_j(\boldsymbol{\omega}_j, \boldsymbol{\Omega}) = \max_{\{I_{f,c}, \{I_{f,j,c}\}_f\}_c} \sum_j q_{f,j} \left(p_{f,j} - c_{f,j}\left(\{I_{f,c}, I_{f,j,c}\}, \boldsymbol{\omega}_f, \boldsymbol{\mathcal{I}}, \boldsymbol{\Omega}\right)\right) - \sum_{c \in C_f} r_c\Big(I_f + \sum_j I_{f,j}\Big) \quad (9)$$

And finally, input prices are determined simultaneously by OEM's take it or leave it offer. OEM's, utilizing their larger market position, choose input prices that minimize their input expenditure subject to the participation of their supplier:

$$p_{f,j} = \arg\min_{p_{f,j}} \underbrace{D(p_j)^{-\sigma} \frac{p_{j,I}}{I_{j,I}^{\alpha}} \frac{p_{f,j}^{-\theta}}{p_{j,I}^{1-\theta}}}_{\equiv q_{f,j}} p_{f,j} \text{ subject to } \tilde{\pi}_{f,j} > 0. \quad (10)$$

The condition $\tilde{\pi}_{f,j} > 0$ is a participation constraint that ensures that supplier f makes positive profits from supplier j accounting for $j's$ contribute to the R&D expenses.[9]

[9]The explicit participation constrain is listed in Appendix Section X. It requires that OEMs compensate suppliers for the entire specialized R&D expenses and their relative share of the generic R&D expenditure.

The stage-1 problem is then to choose the production structure that maximizes the stage 2 profits:

$$\max_{\boldsymbol{\omega}_j} \pi_j\left(\boldsymbol{\omega}_j, \boldsymbol{\Omega}\right) - f\left(\boldsymbol{\omega}_j\right) \text{ and } \max_{\boldsymbol{\omega}_f} \pi_f\left(\boldsymbol{\omega}_f, \boldsymbol{\Omega}\right) - f\left(\boldsymbol{\omega}_j\right).$$

I do not characterize the optimal choices at this point. The estimation will condition on the observes production structure, and counterfactuals will consider exogenous changes in production structure.[10]

Aggregate State. The aggregator function $\boldsymbol{\Omega}$ determines manufacturing proximity from firms' production structures, $\boldsymbol{\Omega} : \{\omega_f, \omega_j\} \mapsto d_{j,I,c}, d_{f,j,c}$. And, the aggregator $\boldsymbol{\mathcal{I}}$determines local R&D stocks from firms' R&D expenditures:

$$\mathcal{I}_c = \sum_f \left(I_{f,c} + \sum_j I_{f,j,c} \right) + \sum_j \left(I_{j,I,c} + I_{j,A,c} \right).$$

[10]In the future, I plan to solve this problem and compute counterfactuals. I additionally want to consider strategic behavior by firms at the point where production structure are determined. That is, compute firms' best-responses to their competitors strategies, anticipating the effect of all production structure on the aggregate state:

$$\max_{\omega_j} \pi_j\left(\omega_j, \boldsymbol{\Omega}\left(\{\omega_{-j}, \omega_f\}\right)\right) - f\left(\omega_j\right) \text{ and } \max_{\omega_f} \pi_f\left(\omega_f, \boldsymbol{\Omega}\left(\{\omega_j, \omega_{-f}\}\right)\right) - f\left(\omega_j\right).$$

0.3 Equilibrium Allocations

In this section I characterized the allocations implied by the solutions to 8, 9, and 10, starting with the optimal input contract.

Lemma 1. *The optimal contract offered by a buyer agrees to purchase inputs at a fixed markup*

$$\mu_{f,j} = 1 + 2\rho\alpha,$$

or, in terms of the optimal price level, a cost-plus contract

$$p_{f,j} = \textit{Unit Cost} + \alpha \times \frac{\textit{R\&D Costs}}{\textit{Quantities}}.$$

The optimal contract is a cost-plus contract that reimburses the suppliers for a multiple α of their R&D investments.[11] The contracted markup is increasing in ρ. When the returns to R&D are high, OEMs allow suppliers to receive greater profits and this incentivize innovation. Additionally, final prices p_j, will be a standard CES markup over marginal costs.

Next, to characterize the R&D allocations, it will be useful to define R&D productivity indices. These indices come from solving out the optimal R&D allocation across countries, allowing us to compactly write other allocation results. For example, with supplier's generic R&D, we use 7 to write

$$\bar{\zeta}_f = \left(\sum_{c \in C_f} \left[\zeta_{f,c} \mathcal{I}_c^{\omega} d_{f,c}^{\tau_f} r_c^{-\zeta} \right]^{\frac{\varsigma}{1-\varsigma}} \right)^{\frac{1-\varsigma}{\varsigma}}. \tag{11}$$

This index embeds the optimal allocation of R&D across countries, and effectively summarizes the marginal cost of raising the level of I_f. Appendix Section X characterizes an equivalent index for specialized R&D $\bar{\zeta}_{f,j}$ and OEM's R&D $\bar{\zeta}_{j,I}, \bar{\zeta}_{j,A}$ using the expressions in 6 and 7. With this notation defined, I present the following allocation results:

Proposition 3. *Upstream firms allocate their R&D across generic and specialized R&D, and within specialized R&D, as follows:*

$$\frac{I_f}{\bar{\zeta}_f} = \sum_j \frac{I_{f,j}}{\bar{\zeta}_{f,j}} \textit{ and } \frac{I_{f,j}}{\sum I_{f,j}} = \frac{\left(\bar{\zeta}_{f,j} q_{f,j} \bar{c}_{f,j}\right)^{\frac{1}{1+\rho}}}{\sum_j \left(\bar{\zeta}_{f,j} q_{f,j} \bar{c}_{f,j}\right)^{\frac{1}{1+\rho}}}.$$

Downstream firms allocate their R&D across components of production according to

$$\frac{I_{j,I}}{I_{j,A} + I_{j,I}} = \frac{\left(\bar{\zeta}_{j,I} p_{j,I}\right)^{\frac{1}{1+\eta}}}{\left(\bar{\zeta}_{j,A} \bar{c}_{j,A}\right)^{\frac{1}{1+\eta}} + \left(\bar{\zeta}_{j,I} p_{j,I}\right)^{\frac{1}{1+\eta}}}$$

[11] I interpret α as reduced-form parameter that captures the bargaining power of suppliers. When α is higher, suppliers can demand greater returns to their R&D investments.

An upstream firms allocates generic R&D expenditures across countries according to

$$\frac{I_{f,c}}{\sum_{c'} I_{f,c'}} = \frac{\left(I_{f,c}\zeta_{f,c}\mathcal{I}_c^{\omega} d_{f,c}^{\tau_f} r_c^{-1}\right)^{\frac{\varsigma}{1-\varsigma}}}{\sum_{c'}\left(I_{f,c'}\zeta_{f,c'}\mathcal{I}_{c'}^{\omega} d_{f,c'}^{\tau_f} r_{c'}^{-1}\right)^{\frac{\varsigma}{1-\varsigma}}}$$

with an equivalent expression for specialized R&D $I_{f,j,c}$ and OEM's allocations $I_{j,I,c}, I_{j,A,c}$.

This characterization highlights four economic forces. First, the division between generic and specialized R&D reflects their relative productivity and returns. When generic R&D is highly productive (high $\bar{\zeta}_f$), suppliers tilt investment towards cost reductions that benefits all buyers. When buyer-specific productivity is high, specialized R&D becomes more attractive.

Second, specialized R&D is allocated towards buyers for which suppliers are productive $\bar{\zeta}_{f,j}$, have high quantity $q_{f,j}$, and have high cost $\bar{c}_{f,j}$. Higher quantities increase the scale over which cost-reducing innovations apply, and higher cost increase the marginal benefit of innovation.[12] When the elasticity ρ increases, the returns to R&D become stronger ad more uniform across buyers.

Third, OEMs allocate R&D towards the components of production where productivity improvements have the highest return - either due to high R&D productivity ($\bar{\zeta}_{j,I}, \bar{\zeta}_{j,A}$) or high production costs ($p_{j,I}$,$\bar{c}_{j,A}$).[13]

Fourth, the cross-country allocation of R&D reflects a standard CES tradeoff between productivity and cost. R&D flows to locations where productivity is high relative to costs. And as $\varsigma \to 1$, expenditure concentrates in the single most productive location.[14]

[12] In equilibrium, higher costs will also reduce the quantities that suppliers receive. Both through competition in input markets, and reduced input purchase from downstream firms. The equilibrium implications of cost shocks should therefore not be interpreted as shocks that necessarily increase innovation.

[13] This result hinges on the assumption that inputs are complements. If inputs were sufficiently substitutable, it would be the case that more R&D would be allocated to the components of production that had the lowest cost. The data supports the fact that inputs are complements.

[14] This theory suggests why directly interpreting reduced-form coefficients from innovation outcomes as causal coefficients, even when using instruments, is problematic. As R&D will endogenously be allocated to the locations where it is most productive.

Table 1: Variable Summary

Variables		Description
Λ_j	:	Share of j in Final Expenditure
$\lambda_{j,A}, \lambda_{j,I}$	:	Share of Assembly/Inputs in j's Cost
$\lambda_{j,f}$	:	Share of Supplier f in Buyer j's Cost
$\lambda_{f,j}$	:	Share of Buyer j in Supplier f's Sales
$\lambda_{f,c}, \lambda_{j,c}$	:	Share of Country c in Firm f's/j's R&D
$\lambda_{c,f}, \lambda_{c,j}$	:	Share of Firm f's/j's in Country c's Total R&D

0.4 Equilibrium Comparative Statics - First Order

To get a better sense of the model mechanism and the potential implications of policy, I characterize the response of industry outcomes GDP to price and R&D efficiency shocks. But changes in prices and R&D efficiency depend on more primitive parameters, and hence we provide some results to think about how prices respond to cost shocks (which affect productivity by raising unit costs) and R&D productivity shocks (which affect the efficiency of R&D investment). Throughout, these comparative statics allow R&D allocations and prices to adjust, but treat ω_f, ω_j as exogenous variables from Stage 1.

I start by noting that changes in real output in this industry can be characterized by the following comparative static:

Proposition 4. *The first-order change in real industry output is given by*

$$d\log GDP = -\sum_j \Lambda_j d\log p_j = -\sum_j \Lambda_j \left(\lambda_{j,A}\left(d\log \bar{c}_{j,A} - \eta\bar{\zeta}_{j,A}\right) + \lambda_{j,I}\left(d\log p_{j,I} - \eta\bar{\zeta}_{j,I}\right)\right),$$

with Λ_j measure the share of firm j in final expenditure, and $\lambda_{j,A}, \lambda_{j,I}$ measure the share of assembly and inputs in the cost of firm j.

Changes in real output arise from either changes in productivity, manifesting as changes in assembly costs $\bar{c}_{j,A}$ or input prices $p_{j,I}$, or changes in R&D efficiency, represented by changes in the indices $\bar{\zeta}_{j,I}$ and $\bar{\zeta}_{j,A}$. While assembly costs are a technological parameter, inputs prices and productivity indices are endogenous through either individual R&D choices or aggregate R&D spillovers.[15] The following results characterize the response of each to primitive cost or R&D productivity shocks.

XXx A partial version of hulten's theorm, but need to acknowledge that input prices are endogenous, and productivity is endogenous

Proposition 5. *The response of input prices to a set of buyer-supplier cost shocks $d\log \bar{c}_{f,j}$ is given by*

$$d\log p_{f,j} = d\log \bar{c}_{f,j} - \sum_k \left(\delta_2 \mathbb{1}_{j=k} + \delta_3 \lambda_{f,k}\right)\left((1-\theta)d\log \bar{c}_{f,k} + \tilde{\lambda}_{j,I} d\log p_{k,I}\right) - \delta_1 \sum_k \Lambda_k \lambda_{k,I} d\log p_{k,I}$$

where $d\log p_{k,I} = \sum \lambda_{k,f} d\log p_{f,k}$. The variable $\lambda_{f,j}$ is the share of downstream firm j in upstream firm f's revenue, and $\lambda_{j,f}$ is the share of upstream firm f in the input price of downstream firm j. The variable

[15] Assembly costs are a technological parameter as we have conditioned on the Stage 1 choices. In principle, these too are endogenous to the plant location and market entry choices of firms.

$\mathbb{1}_{j=k}$ is an indicator that takes a value of 1 when buyer k in the summand equals buyer j. The parameters δ_1, δ_2, and $\delta_3$are defined below.

This comparative static reflects several forces. First, there is the direct effect of costs to prices, which ignoring any adjustments in quantity or R&D, directly pass through due to the pricing implications of Lemma 1.

Second, for every buyer in a supplier's network, there is a potential supplier-specific cost shock $d\log\bar{c}_{f,k}$, which affects R&D incentives as it increases costs and lowers demand (hence the $1-\theta$). And, each buyer is also experience a change in its total input costs $d\log p_{k,I}$, arising from changes in the prices offered by each supplier. Total input costs affect the R&D incentives of suppliers through four separate mechanisms:

$$\tilde{\lambda}_{j,I} \equiv \underbrace{\theta-1}_{\text{Upstream Competition}} + \underbrace{1}_{\text{Input Expenditure}} - \underbrace{\frac{\rho}{\rho+1}}_{\text{Input R\&D}} - \underbrace{\frac{1}{\rho+1}\frac{\sigma}{1+\rho(1-\sigma)}\lambda_{j,I}}_{\text{Downstream Demand}}.$$

Higher input prices faced by buyer j, benefit each supplier as it perceives weaker perceive competition and higher input expenditure. However, higher input prices lead downstream firms to reallocate R&D towards inputs and raise final prices, which both lower input demand. The resulting changes in R&D affect costs through though the reallocation of specialized R&D, which affects particular buyers, and expansion of generic R&D spending as well as the overall R&D budget, which benefits all buyers:

$$\delta_2\mathbb{1}_{j=k}+\delta_3\lambda_{f,k} = \underbrace{\frac{\rho}{1+\rho(1-\theta)}}_{\text{Reallocation of R\&D to } j}\times\mathbb{1}_{j=k}+\left(\underbrace{\frac{2\rho}{1-2\rho(\theta-1)}}_{\text{Generic R\&D and R\&D Budget}} - \underbrace{\frac{\rho}{1-\rho(\theta-1)}}_{\text{Reallocation of R\&D to } k}\right)\times\lambda_{f,k}.$$

The final term normalizes changes in $d\log p_{k,I}$ to emphasize that relative prices changes are what determines reallocation in downstream demand: $\delta_1=(\delta_2+\delta_3)(\sigma-1)/1+\rho(1-\sigma)$. But notably, this last term will not cancel out the other $d\log p_{k,I}$ terms since even if every firm experienced the same cost shock such that relative quantities did not change, the absolute level of demand would change which would affect the R&D incentives of firms.

[XXX] Matrix formulation - then what do we need for uniqueness?

Specializing Proposition 5 to a single buyer, removes terms associated with reallocation across buyers, and highlights the key sources of inefficiency in the model.

Corollary 1. *Consider the case with only one downstream buyer, the change in real GDP can be expressed as*

$$d\log GDP = -\frac{1}{1-2\rho\left(\frac{1}{1+\eta}\lambda_{j,I}+\frac{\eta}{1+\eta}\right)}\sum_f \lambda_{j,f}\, d\log c_{f,j}$$

This result echo's Hulten's theorem, but has a non-unitary coefficient. This arises due to the presence of a baseline externality, alluded to in Proposition 1 from the failure of Shephard's Lemma. Suppliers do not capture the full extent of their productivity improvement. In particular, they do not capture the fact that productivity improvements affect downstream quantities, $\frac{1}{1+\eta}\lambda_{j,I}$, and downstream R&D allocation, $\frac{\eta}{1+\eta}$. Notably, when there is no R&D and $\rho=0$, we revert to the standard case.

[XXX] Under what conditions does this equal 1? Is it less than 1? Or is it greater than 1?

Next, I discuss changes in R&D efficiency:

Proposition 6. *Consider the response of the productivity index $\bar{\zeta}_f$ to a set of shocks $\{\zeta_{f,c}\}\,\forall f, c$:*

$$d\log\bar{\zeta}_f = \sum_{c\in C_f}\lambda_{f,c}\Bigg(\underbrace{d\log\zeta_{f,c}}_{\textit{Direct Productivity Effect}} + \frac{\omega}{1-\varsigma}\sum_{f'}\lambda_{c,f'}\underbrace{\left(d\log\zeta_{f',c} + d\log K_c - \sum_{c'\in C_{f'}}\lambda_{f',c'}\left(d\log\zeta_{f',c'} + d\log K_{c'}\right)\right)}_{\textit{Spillovers from Across-Country R\&D Reallocation}}$$
$$- \omega\underbrace{\left(\frac{1}{\rho}\sum_{f'}\lambda_{c,f'}\sum_{j}\lambda_{f',j}d\log p_{f',j} + \frac{1}{\eta}\sum_{j}\lambda_{c,j}d\log p_j\right)}_{\textit{Spillovers from R\&D Expansion}}\Bigg)$$

where $\lambda_{f,c}$ is the share of R&D expenditure by firm f in country c, $\lambda_{c,f'}$ is the share of R&D in country c done by firm f'.

This comparative static reflects several forces. First, the R&D efficiency index responds to R&D productivity shocks in all countries, with shares depending on the extent of baseline R&D expenditure done in each country. Second, R&D efficiency responds to R&D shocks at all firms and all countries, to the extent that the strength of R&D spillovers measured by ω are higher. In particular, based on R&D productivity shocks, firms relocate their R&D expenditure across countries - where the higher substantially there is ς, the stronger these effects are. If other firms are relocating R&D to the countries where firm f does R&D intensely, then this firm benefits. Finally, in general, R&D productivity changes will induce more R&D investment by suppliers and buyers. In the absence of cost-shocks, prices only adjust due to R&D changes and hence scaling prices changes by the inverse of the returns to R&D get the expansion in total R&D. Obviously, prices are a potentially complicated endogenous object, so we characterize their response to R&D shocks in Appendix Section X.

Corollary 2. *We have*

$$d\log GDP = -\frac{\rho}{\left(1+2\rho-2\rho\left(\frac{\eta}{1+\eta}+\frac{1}{1+\eta}\lambda_{j,I}\right)-\rho\omega\left(\frac{1}{\eta}\lambda_{c,j}+\frac{1}{2\rho}\lambda_{c,f}\right)\right)}\lambda_{j,I}d\log\zeta_{f,c}$$

where $\lambda_{c,j}$ is the share of R&D conducted by buyers, and $\lambda_{c,f}$ is the share of R&D conducted by suppliers.

0.5 Equilibrium Comparative Statics - Second-Order

The final result we provide in this section is a characterization of the second-order effect of cost shocks. To parse these results, I first introduce some notation. First, we use Proposition 5 to characterize the first-order price response to cost changes, which we compactly represent as

$$\psi^j_{f,k} = d\log P_j / d\log c_{f,k}.$$

Second, we introduce two notions of covariance in the first-order price response. The first of these,

$$Cov_\Lambda\left(\lambda_{j,I}\psi^j_{f,j}, \lambda_{j,I}\psi^j_{f,k}\right) = \sum_j \Lambda_j\lambda_{j,I}\psi^j_{f,k}\lambda_{j,I}\psi^j_{f,j} - \sum_j \Lambda_j\lambda_{j,I}\psi^j_{f,j}\sum_k \lambda_{k,I}\Lambda_k\psi^j_{f,k},$$

measures the covariance in prices changes experienced by downstream buyers, where we use the distribution of buyers in the downstream price index Λ_j. The second of these,

$$Cov_{\lambda_f}\left(\tilde{\lambda}_{j,I}\psi^j_{f,j}, \tilde{\lambda}_{j,I}\psi^j_{f,k}\right) = \sum_j \lambda_{f,j}\tilde{\lambda}_{j,I}\psi^j_{f,j}\tilde{\lambda}_{j,I}\psi^j_{f,k} - \sum_j \lambda_{f,j}\tilde{\lambda}_{j,I}\psi^j_{f,j}\sum_k \lambda_{f,k}\tilde{\lambda}_{k,I}\psi^l_{f,k}.$$

again measures the covariance in prices changes experienced by downstream buyers, but instead uses the distribution of buyers in the profits of supplier f. Using these definitions. We can now characterize the response of second-order effect of cost shocks on input prices.

Proposition 7. *The second-order response of input prices to a set of buyer-supplier cost shocks $\{d\log\bar{c}_{f,j}, d\log\bar{c}_{f,k}\}$ where $j \neq k$ is given by*

$$\begin{aligned}\frac{d^2\log p_{f,j}}{d\log\bar{c}_{f,k}} =& \delta_8\sum_k\left(\delta_2 \nVdash_{j=k} + \delta_3\lambda_{f,k}\right)\lambda_{j,I}\left(1-\lambda_{j,I}\right)\psi^j_{f,k}\psi^j_{f,j} + \delta_7\delta_1\sum_j \Lambda_j\lambda_{j,I}\left(1-\lambda_{j,I}\right)\psi^j_{f,k}\psi^j_{f,j} \\ &- \delta_5 Cov_{\lambda_f}\left(\lambda_{j,I}\psi^j_{f,j}, \lambda_{j,I}\psi^j_{f,k}\right) + \delta_1\delta_6 Cov_{\Lambda}\left(\lambda_{j,I}\psi^j_{f,j}, \lambda_{j,I}\psi^j_{f,k}\right)\end{aligned}.$$

Now to unpack this expression, we define each new parameter in turn. First, recall from above that δ_1 determines how changes in the aggregate downstream price index affect the profit incentive for innovating for each buyer. With that in mind, note that

$$\delta_6 = \frac{\sigma-1}{1+\rho\left(1-\sigma\right)},$$

which is the elasticity of downstream demand shares with respect to cost changes. What this term captures is how the first-order changes in the final prices of downstream firms relocate quantity across buyers. These quantities in turn affect the demand faced by various suppliers and thus their incentive to do R&D. The covariance appears as a measure of the extent to which buyers which are experiencing large first-order price effects are in the second-order also experiencing reallocations of demand shares towards them. Relatedly, we have $\delta_7 = 1/\left(1+\rho\right)$, which captures how for each downstream firm in the final price index, the cost share of inputs is changing - and hence the ability of upstream R&D to influence downstream prices.

Next, recall that δ_2 and δ_3 determine how changes in the input prices of various buyers affects the R&D incentives of suppliers either through a competition effect or a reallocation/expansion of input demand. Now note,

$$\delta_{4,j} = \frac{1}{\rho+1},$$

which similar to δ_7, captures how within each buyer j the share of costs that is determined by upstream R&D choice is changing, and this amplifies any first-order feedback effect between upstream R&D and demand for inputs. The first component of $\delta_{4,j}$ captures how this second-order effect does not alter the competition effect from the change in input prices which is governed by $-\theta$; it only influences the input demand expansion channel $\frac{1}{\rho+1}\frac{\sigma}{1+\rho(1-\sigma)}\lambda_{j,I}$.

Finally, we have another change stemming from the common component of upstream R&D, where we have

$$\delta_5 = \frac{1}{1-\rho\left(\theta-1\right)},$$

which is the elasticity of profit shares $\lambda_{f,j}$ with respect to profit shifters y_j. And why this appears with a covariance term is to measure the extent to which buyers which are experiencing large first-order price effects are in the second-order also experiencing reallocations of profits shares in supplier f toward them.

XXX Note how this $\bar{c}_{f,j}$ can also be a wedge. Policy instrument - effect on outcomes.

$$\delta_8 = \frac{-\sigma}{1+\eta\left(1-\sigma\right)}\frac{1}{\left(1+\eta\right)^2}$$

[X] Comparativ Static - Second Order Productivity Change

0.6 Specialized Equilibrium Characterization

[X] are offshoring choices by firms complements or subsitties? Look at effect on horizontally linked firms, and vertically linked firms, business stealling, veruss shared supplier network!, versus local spillovers ... versus market expansion?

[X] Comparative Statics - Can go here next if prev is finished.

[X] Firm Expansion and growth - additional of new location?

[X] Is there a model I can write with entry? Probabilistic sort of like https://drive.google.com/file/d/15sBES8Eh6LYARw4PaVfzbmP/view?

SO there is a firm, they hire a certain amount in a location, and they innovate a certain amount. The probabilistically enter each market.

[X] Since prices not prop to R&D, are these distributional gains from reallocating demand across firms - generate more R&D? More spillovers? Esp with diminishing returns?

1 Symmetric Industry Equilibrium

The previous section highlighted how the structure of a production network generates a dependence between the R&D choices of firms vertically linked as suppliers and buyers, and those horizontally linked either through competition or a shared set of suppliers. While the heterogeneity present in that model is key for any quantitative statements, one can push the theory and develop further intuition by focusing on a simplified case. In this section I characterize an industry equilibria with homogenous upstream and downstream firms - abstracting from any heterogeneity in technology or geography. This simplification aims to make transparent how the model parameters govern the response of the economy to outsourcing and offshoring shocks.

Specifically, I assume there is a homogenous mass M of suppliers and mass N of buyers, all innovating in the same location. Additionally, I assume that there is an entry cost for upstream and downstream firms that is convex in the number of entrants:

$$\pi_f^{\gamma_f} = M f_M \quad \text{and} \quad \pi_j^{\gamma_j} = N f_N,$$

where $\gamma_f, \gamma_j \in [0, 1]$.[16]This entry process along with a homogenous versions of the previous equations, completes the characterization of the industry equilibrium.

Definition. A symmetry equilibrium is characterized by upstream prices and quantities $(p_{f,j}, \tilde{x}_j)$, profits (π_f, π_j), entrants (M, N), and spillovers (K) that satisfy Appendix Equations X.

I now provide two lemmas that I draw on and follow with comparative statics for the industry equilibrium. I will only be considering comparative statics with respect to distances and the exogenous component of costs, which is a necessary simplification for these results.

Lemma 2. *In a symmetric equilibrium, there is no change in the number of downstream firms in response to a change in production costs or distance. Additionally, the total R&D expenditure of downstream firms remains fixed:*

$$d \log N = d \log \pi_j = d \log \left(I_{j,I} + I_{j,A} \right) = 0.$$

This result largely follows from the assumption of fixed market size and a fixed markup. If we allowed substitution with an outside sector, e.g. $D = \bar{D} P^{\sigma - 1 - \kappa}$ through quasi-linear preference, then this would not continue to hold. Changes in costs would create a market expansion effect, which would result in room for more downstream entrants.

Lemma 3. *In a symmetric equilibrium, in response to a change in production costs or distance, we can express the change in profits as directly related to the change in input prices:*

$$\left(1 + (1 + \gamma_f) \lambda_U \frac{\omega}{\rho} \right) d \log \pi_f = -\frac{1}{2\rho} d \log p_{f,j} + \frac{\tau}{2\rho} d \log d_f + \frac{\tau}{2\rho} d \log d_{f,j},$$

and in turn we can relate

$$\frac{d \log K}{d \log \pi_f} = (1 + \gamma_f) \lambda_U \ \textit{and} \ \frac{d \log M}{d \log \pi_f} = \gamma_f$$

where λ_U is the contribution of upstream firms to knowledge spillovers.

[16]One can imagine a factor in fixed supply - like land in the industry cluster - that generates these convexities in entry.

This is a symmetric equilibrium version of Proposition **??**. The inverse elasticity of prices with respect to R&D is $1/2\rho$, and since profits are proportional to R&D expenditure, we can use changes in upstream prices as a sufficient statistic for profits and thus entry. Finally, this also lets us extrapolate to changes in the local knowledge stock K since again R&D is proportional to profits, and it is only upstream profits which are changing - following Lemma 2.

Finally, I provide sufficient conditions under which an equilibrium exists and is unique. One reason we might be worried about uniqueness or existence in this model is the external knowledge spillovers in K, and issues related to the R&D coordination between upstream and downstream firms.[17]

Proposition 8. *A sufficient condition for equilibrium existence and uniqueness when $\omega > 0$ is*

$$\frac{2\omega}{1+2\rho}\left(\left(1+\gamma\right)\sigma_{\pi,p}\right) + \frac{2\rho}{1+2\rho}\left[-\gamma\sigma_{\pi,p} + \left(1+\frac{1}{\theta}\gamma\sigma_{\pi,p}\right)\right] < 0$$

where $\sigma_{\pi,p} = 1/2\rho\left(\left(1+\gamma\right)\frac{\omega}{p}+1\right)$. Or, if $\omega = 0$, we then have

$$\underbrace{\frac{-\gamma}{2\rho}}_{\textit{Business Stealing}} + \underbrace{\left(1+\frac{1}{\theta}\frac{\gamma}{2\rho}\right)}_{\textit{Market Expansion}} < 0.$$

There are several forces that work to determine equilibrium existence and uniqueness. To walk through them, imagine taking input prices to zero. For existence, we want to make sure the changes that occur to firm R&D incentives along this price path don't support an equilibrium with a price of zero. Following Lemma 3, this price change is associated with higher profits, more entry, and greater knowledge spillovers. How do these changes in turn affect the R&D incentives of firms? The knowledge spillovers are consistent with lower prices. And the lower prices themselves along with the lover of variety effect from entry are generating higher input demand which is consistent with high R&D and low price. Existence then hinges on a business stealing effect, which has to be sufficiently strong such that the incentive for R&D vanishes as prices go to zero. A similar logic works for taking prices to infinity.[18]

1.1 Industry Planner Solution

I start by relating the solution from the decentralized market equilibrium to a pseudo social-planner. Namely, I consider an industry planner which coordinates the investment of upstream and downstream firms to maximize industry profits, while facing an iso-elastic demand curv e with price elasticity $-\sigma$. For instance, imagine a national industry planner trying to coordinate behavior in an industry, hoping to win global market share whilst facing an elasticity of substitution across countries of σ.

Proposition 9. *An industry planner would choose an allocation of R&D for input technology across down-*

[17] XXX Big-push type equilibria where upstream and downstream firms have to collectively undertake R&D to move into a higher profit equilibrium. Neither has the private incentive to do it alone.

[18] XXX Note although these are sufficient conditions, they ought in practice be close to the necessary conditions. The proof relies on bounding an additive component by its upper bound.

stream and upstream firms that satisfies

$$1 = \frac{\sum_f I_f^{Planner}}{I_{j,I}^{Planner}}, \quad \frac{(\theta-1)(\mu-1)}{\mu M^{-1/\theta}} = \frac{\sum_f I_f^{Decentralized}}{I_{j,I}^{Decentralized}}.$$

And, if firms additionally differ in the exogenous component of their costs, a planner would allocate

$$\frac{\left(\bar{c}_{f',j}\right)^{\frac{\theta}{1-2\theta\rho}}}{\left(\bar{c}_{f,j}\right)^{\frac{\theta}{1-2\theta\rho}}} = \frac{I_f^{Planner}}{I_{f'}^{Planner}}, \text{ and } \frac{\left(\bar{c}_{f',j}\right)^{\frac{\theta-1}{1-2(\theta-1)\rho}}}{\left(\bar{c}_{f,j}\right)^{\frac{\theta-1}{1-2(\theta-1)\rho}}} = \frac{I_f^{Decentralized}}{I_{f'}^{Decentralized}}$$

For planner that optimizes across firms:

Looking at equations **??** and **??**, we can see the elasticity of input costs with respect to upstream or downstream input R&D is ρ. Since these firms additionally face the same costs, it follows that the input R&D should be equivalent between upstream and downstream sectors in a planned economy. However, when we move to the decentralized allocation, the private returns to input R&D will differ across stages. First, markups have two offsetting effects - higher markups lead to more R&D upstream, but also raise the input cost leading downstream firms to compensate with their own R&D. Second, we have the $\theta - 1$ term, which is the elasticity of profits with respect to R&D, a similar R&D profit motive is absent in the planned economy. Finally, there is additionally over investment upstream due to a love-of-variety effect that is not internalized by any upstream firms.

XXX Who to subsidize for spillovers. Bosch might have a production network that supports a lot of competitors, but it might also generate large local spillovers!

This suggests that downstream firms enjoy an environment with low variance in preference heterogeneity (i.e. high θ). While we take the preference distribution as exogenous, one can imagine that if firms create increasingly homogenous downstream products or if entrants mimic existing products, then their preferences for inputs will become similar. This homogeneity in demand will lead to more excess upstream competition (relative to the preference of an industry planner), which moves the incidence of input R&D onto the balance sheet of suppliers.

Finally, while we focus here on a homogenous firm model, it is worth noting that if there is heterogeneity in suppliers, there will be underinvestment by the lower cost suppliers. This is because while cost reductions achieved through R&D lead to higher demand, they lower the price received on those units of production. As for the industry planner, since any markup is just a transfer in profits between suppliers and buyers, the planner does not have this offsetting motive from lower prices. Note, we have a simplified model setup with fixed markups, if markups are higher for lower cost firms, then some of this underinvestment will be offset.

The market allocation is clearly not efficient, and this simplified model doesn't touch on other forces that would be present in a heterogenous firms model like business stealing or the input market expansion effects that might generate further wedges from the first-best. One obvious implication of this, there is room for policy to have first-order positive effects on welfare.

1.2 Sourcing Inputs from Abroad

The first change we consider is a change in the price of foreign inputs p_{Foreign} - perhaps spurred by tariff changes or a technology shock abroad. To implement this change, we write the input cost of downstream firms as $P^{-\theta} = (Mp_{f,j}^{-\theta} + p_{\text{Foreign}}^{-\theta})$, where the model accounts for endogenous adjustments in M and $p_{f,j}$ but takes the foreign price as exogenous. We start by characterizing the effect on input prices

Proposition 10. *A change in the price of foreign inputs has an equilibrium effect on domestic input prices of*

$$(1-\eta_a)\frac{d\log p_{f,j}}{d\log p_{Foreign}} = \frac{-2\rho}{1+2\rho(1-\theta)}\left(\underbrace{\theta}_{Competition} + \underbrace{\frac{-\rho}{\rho+1} - \frac{1}{\rho+1}\lambda_j^{p_j^F}}_{Expansion\ of\ Input\ Demand}\right)\lambda_{Foreign}$$

where η_a, defined in Appendix Equation X, summarizes the feedback effect of input prices on itself which includes the effects on quantities, entry, and the stock of local knowledge.

The first direct effect of a change in the foreign input price is one of competition. Domestic suppliers now receive a smaller share of the total input demand and as a result face less incentive to innovate. At the same time, the decrease in input prices faced by downstream firms leads them to demand more inputs and adjusts their R&D expenditure away from inputs - freeing up more R&D to be done by suppliers. On balance, since $\theta > 1$, the competition effect dominates, and tougher foreign competition leads to higher input prices.[19]

Now to extrapolate to the effect on final prices, we need to account for a few more changes. Using Lemma 3, we note that alongside the change in input price, we have a change in the number of upstream entrants and a change in the stock of local knowledge. The change in the number of entrants of course has effects on the input price, contained in η_a, but they also have first-order effects in final prices through the love-of-variety effect and the effect of knowledge spillovers on the productivity of R&D by downstream firms. So, even if total expenditure on inputs falls, if foreign sourcing sufficiently deteriorates the amount of local R&D, then it can result in higher final prices.

$$\frac{d\log p^F}{d\log p_{\text{Foreign}}} = \underbrace{\frac{d\log p_{f,j}}{d\log p_{\text{Foreign}}}}_{<0}\left(\lambda_{\text{Home}}\lambda_j^{p_j^F}\left(1-\frac{1}{\theta}\underbrace{\frac{d\log M}{d\log p_{f,j}}}_{<0}\right) - \omega\underbrace{\frac{d\log K}{d\log p_{f,j}}}_{<0}\right) + \lambda_{\text{Foreign}}\lambda_j^{p_j^F}$$

So if a downstream firm determined that foreign sourcing, while lowering input costs, could lead to ultimately worse local conditions, then what other strategies could it pursue? For one, it could set up a joint venture with the input supplier to move its R&D investment or knowledge from abroad into the local economy. This is perhaps what we are seeing in the partnerships between legacy automakers in the US and battery suppliers from China, Japan, and South Korea - joint ventures that seek to lower input prices but also transfer technology and know-how into the local economy. Additionally, note that we abstracted from endogenous adjustments abroad. A knock-on effect from foreign sourcing is that foreign suppliers will now undertake the input R&D previously done by local firms. Presumably those foreign firms transact more with

[19] XXX Need to connect to literature on escape competition. How do you do this? Do you need to have firms intenalize effect on price index.

foreign downstream competitors and share local knowledge spillovers with those firms. The reallocation of R&D abroad would then worsen the relative technological position of domestic firms.

Finally, some caveats to these results. Note again that Proposition 10 suggests that prices offered by domestic firms can only rise in response to foreign competition. We know that empirically, this isn't always the case (XXX reference), so here we offer three caveats to the model that allow foreign sourcing to lead to greater levels of innovation by domestic first. First, note that in a model with heterogeneous firms, the firms exiting would be the lowest productivity ones. This may lead to reallocation of demand such that remaining high productivity firms experience a rise in R&D. In light of Proposition 9, this reallocation corrects for underinvestment by high productivity firms.

Market Expansion Second, the baseline model keeps the extent of the market for downstream goods fixed. But if this were not the case, if aggregate demand increased as prices fell, then foreign sourcing could lead to a more demand for all input suppliers. Suppose that the demand shifter satisfied $D = \bar{D}P^{\sigma-1-\kappa}$, then Proposition 10 could be adjusted as follows:

$$\text{Expansion of Input Demand:} \quad \frac{-\rho}{\rho+1} - \frac{1}{\rho+1}\lambda_j^{p_j^F} \Rightarrow \frac{-\rho}{\rho+1} - \frac{1}{\rho+1}\frac{1+\kappa}{1-\kappa\rho}\lambda_j^{p_j^F}$$

where if κ is sufficiently large, then the market expansion effect of foreign sourcing would lead to gains for even domestic suppliers.

Escape Competition Third, suppose domestic inputs were supplied by an oligopolist f that internalizes its impact on the supplier price index when it determines its market share for the input demand from the buyer.[20] Their new optimization problem is then:

$$\max_{I_f} \tilde{x}_j \frac{\mu-1}{\mu} \frac{\left(\mu\bar{c}_{f,j}I_f^{-2\rho}\right)^{1-\theta}}{\left(\mu\bar{c}_{f,j}I_f^{-2\rho}\right)^{-\theta} + \left(p_{\text{Foreign}}\right)^{-\theta}} - 2rI_f$$

The firm's optimal R&D spending then satisfies

$$I_f^{\rho} = \left(\left(\rho\left(\theta-1\right)\lambda_f - \rho\left(\theta\right)\lambda_f^2\right)\frac{\tilde{x}_j}{r}\right)^{\frac{1}{2}}$$

where $\lambda_f = (\mu\bar{c}_{f,j}I_f^{-\rho})^{-\theta}/((\mu\bar{c}_{f,j}I_f^{-\rho})^{-\theta} + (p_{\text{Foreign}})^{-\theta})$. In the limiting case where $p_{\text{Foreign}} \to \infty$, then there is no incentive to invest as suppliers already capture the whole market and cost reduction will only lower the price they charge. A shock from foreign competition can lead to more innovation as firms now see a reason to spend on R&D to win back market share. In Figure 1, I plot R&D as a function of the foreign price. Note the non-monotonic response. Decreases in price can either lower domestic R&D as they take market share and potential profits away. Or, price decreases can raise domestic R&D as they undermine the monopoly position of the domestic firm and give it reason to invest to win back market share.

[20] I continue to assume that the firm does not internalize its impact on downstream prices and hence aggregate input demand.

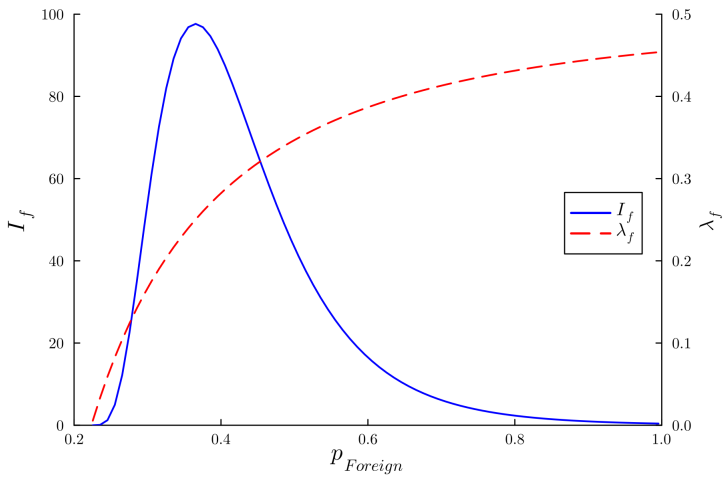

Figure 1: Innovation to Escape Foreign Competition

1.3 Offshoring Assembly Abroad

I next consider how offshoring assembly impacts input prices through the endogenous adjustment of R&D. While, we have abstracted from a discrete sense of geography in this industry equilibrium, we can approximate assembly offshoring through three changes. First there is a decrease in assembly costs, $c_{j,A} \downarrow$, second an increase in the distance between downstream innovation and assembly $d_{j,A}$, and third an increase in the distance between upstream innovation and downstream production $d_{f,j}$. The following proposition summarizes the first-order effects of these three changes on input prices.

Proposition 11. *Offshoring assembly has an equilibrium effect on input prices of*

$$(1-\eta_b)\, d\log p_{f,j} = \frac{\tau}{1+2\rho} d\log d_{f,j} + \frac{2\rho}{1+2\rho}\frac{1}{\rho+1}\left(1-\lambda_j^I\right)\left(d\log c_{j,A} + \tau d\log d_{j,A}\right)$$

where η_b, defined in Appendix Equation X, summarizes the feedback effects of input prices on itself which includes the effects on quantities, spillovers, and entry.

A decrease in assembly costs or increase in assembly R&D efficiency (e.g. decrease in $d_{j,A}$) reduces prices and hence increases physical quantities. This raises the demand for inputs, captured by $(1-\lambda_j^I)/(1+\rho)$, which in turn raises the R&D incentives of upstream firms and hence lowers prices (R&D response on prices is captured by $2\rho/(1+2\rho)$).[21] A more direct effect is the effect of upstream R&D efficiency, i.e. the rise in $d_{f,j}$, which has a first-order effect on prices.

Two notes on Proposition 11. First, it is not clear without estimate of τ and ρ what the effect of offshoring is in input prices in a symmetric equilibrium - this is because $c_{j,A}$ and $d_{j,A}$ move in opposite directions - offshoring offers better labor conditions for production but makes innovation harder due to the geographic fragmentation of production. Second, even if the effect of offshoring was to raise input prices,

[21] The $(1-\lambda_j^I)/(1+\rho)$ contains two forces - both the effect of lower prices on higher quantities - but also the lower assembly prices in particular leads to downstream R&D reallocation towards inputs, which dampens the quantity effect.

final prices might still fall. The determination follows from the following calculation:

$$d\log p^F = \left(1-\lambda_j^I\right)\left(d\log c_A + \tau d\log d_{j,A}\right) + d\log p_{f,j}\left(\lambda_j^I\left(1-\frac{1}{\theta}\frac{d\log M}{d\log p_{f,j}}\right) - \omega\frac{d\log K}{d\log p_{f,j}}\right)$$

More concretely, changes in input prices will only alter final prices according to λ_j^I - their share in final prices. If this share is small, or if the rise in input prices is insignificant, offshoring may still lower final prices. But, outside of the direct effect of an input price change, offshoring - if it raises input prices - will also lead to upstream exit and will lead to lower R&D investment by suppliers - deteriorating the spillover component of innovation.

One caveat with this result, no firm uses the response in Proposition 11 to calculate their returns to offshoring - this is purely the equilibrium effect where all firms experience the same change in fundamentals. To get some sense of the private incentives, we next consider the change in input prices faced by a particular infinitesimal firm when it alone offshores.

Proposition 12. *Consider offshoring by an infinitesimally small firm which poses no changes on prices indices or the aggregate demand faced by suppliers:*

$$\left(1-\eta_c\right)d\log p_{f,j} = \frac{\tau}{1+\rho\left(1-\theta\right)}d\log d_{f,j} + \frac{\rho}{1+\rho\left(1-\theta\right)}\frac{\sigma}{1+\rho\left(1-\sigma\right)}\frac{1}{1+\rho}\left(1-\lambda_j^{p_j^F}\right)\left(d\log c_{j,A} + \tau d\log d_{j,A}\right)$$

where η_c, defined in Appendix Equation X, summarizes the feedback effects of input prices on itself which includes the effects on quantities, spillovers, and entry.

This expression offers several differences; to start, the $2\rho/(1+2\rho)$ is replaced by two terms. First, there is the fact that a firm that offshores has a lower relative price - hence higher demand. This is a business stealing effect - where the elasticity of substitution across downstream firms dictates the returns to offshoring. Second, there is the fact that when faced with greater demand from a particular buyer, upstream firms will invest more for that buyer, and extent of the investment response on the profit elasticity $\rho\left(\theta-1\right)$. It is also clear from the fact that $\theta > 1$ and $\sigma > 1$, that the private returns from unilateral offshoring are larger than the equilibrium returns. Combined with fixed costs of offshoring, there is a possibility for a prisoner's dilemma to emerge. It may be privately optimal for any given firm to offshore from an equilibrium where all firms are on-shore, but the equilibrium achieved when all firms offshore leaves everyone worse off.

The Input Price Feedback Loop Finally, I unpack the term η_b, while there are subtle differences in η_a, η_b, η_c - they all take the same general structure and capture the ways a change in input prices feedback on itself through equilibrium adjustments like entry, demand, and spillovers:

$$1-\eta_b \equiv 1 + \frac{2\rho}{1+2\rho}\left[\underbrace{\left(\frac{-\rho}{\rho+1} - \frac{1}{\rho+1}\lambda_{j,I}\right)}_{\text{Reallocation and Scale Effect on } \tilde{x}}\left[1-\underbrace{\frac{1}{\theta}\frac{d\log M}{d\log p_{f,j}}}_{\text{Love of Variety}}\right]\right] + \underbrace{\frac{2\omega}{1+2\rho}\frac{d\log K}{d\log p_{f,j}}}_{\text{Spillovers}} - \underbrace{\frac{2\rho}{1+2\rho}\frac{d\log M}{d\log p_{f,j}}}_{\text{Business Stealling}}$$

Where recall that Lemma 3 provides expressions for $d\log M/d\log p_{f,j}$ and $d\log K/d\log p_{f,j}$ in terms of model parameters. As for intuition, we first have the downstream R&D reallocation and scale effects.

Higher input prices lead to more downstream R&D on inputs, and less downstream quantities which both reduce the demand faced by upstream firms. Where note, the price faced by downstream firms for inputs consists of both the input price itself and the love of variety effect from the number of suppliers. Additionally, input prices affect entry and local R&D spillovers - with spillovers reducing prices by raising the efficiency of R&D and entry raising prices through a business stealing effect where each firm receives a lower share of demand and hence faces a lower return to cost reductions.

XXXX This is just taking the matrix inverse and collapsing it since we have homogenous firms.

XXXX Is it then possible to stick with the matrix stuff, by do a 2-country case!!!! So like can one analytically use the matrix inversion? What assumption would need to be made to simplify things!!!

XXXXX Optimal policy - should it be a tariff or a subsidy? Well the tariff is coordination device - don't actually need to pay out anything. But the subsidy you actually have to pay out! So which one is optimal?

1.4 Ideas

XXXX I need a theory with multi-locations.

XXXX Do that next. Have entry that is country-specific? Will that be tractable?

XXXX What if spillovers are input-specific? So like there are spillovers from only input R&D onto supplier R&D, then that substition matters a lot.

XXXX Nash in production network. Assume homogenous ? Assume that downstream firms anticipate price response of upstream.

XXXX If there is relative underinvestment by high productivity in $I_{f,j}$, what does that mean about $I_f/$

XXXX What if there are heteogenous innovation costs? Should the buyer, if they have lower innovation costs, over-innovate?

XXXX Compute Nash numerically!!!

XXXX You can think of input innovation as in a reduced-form way capturing outsourcing!!! Firms innovate to reduce dependence on outside firms, instead supplying some of the input internally.

XXXX In a monopolistic competition model, profits are prop to revenue. Revenues are fixed in this model with insdustry demand fixed. So no adjustment in downstream R&D.

XXX Dynamic model. So everywhere in my model, it has to be the case the the marginal value of innovation is equalized. Each firm is setting marginal profit equal to marginal cost, hence it must be the same so long as marginal costs are the same.

XXXX Would a recursive structure help?

$$V_j = \lambda_1 D^{\frac{1}{1+\rho(1-\sigma)}} \left((c_A K_A)^{\frac{1}{\rho+1}} + (c_I K_I)^{\frac{1}{\rho+1}} \right)^{\frac{(\rho+1)(1-\sigma)}{1+\rho(1-\sigma)}} + V_j$$

[**XXX Dynamic Model**] [XXXX Dynamic model with growth in demand]